

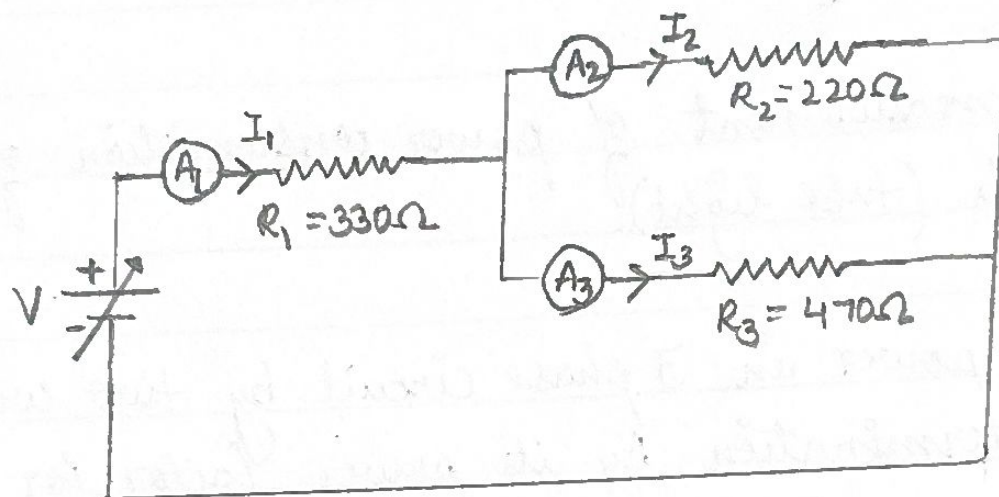


Fig-1: Circuit Diagram of KCL

Object : Verification of Kirchhoff's Law

Apparatus Required :

S.No.	Equipment Name	Specification	Quantity
1.	Isolated variable DC voltage Source	Digital	
2.	Voltmeter		
3.	Ammeter		
4.	Fuse for short circuit protection		
5.	Patch cords.		

Brief Theory :

Kirchhoff's Current Law (KCL):

Kirchhoff's first law is based on the Law of Conservation of charge, which requires that the algebraic sum of charges within a system cannot change.

It states that the algebraic sum of current of a node (junction) in a network at any instant is zero.

KCL can be expressed mathematically as $\sum I = 0$

There are three currents at node 'n' in Fig.1 Applying KCL at Node 'n'

$$i_1 + (-i_2) + (-i_3) = 0$$

$$i_1 = i_2 + i_3$$

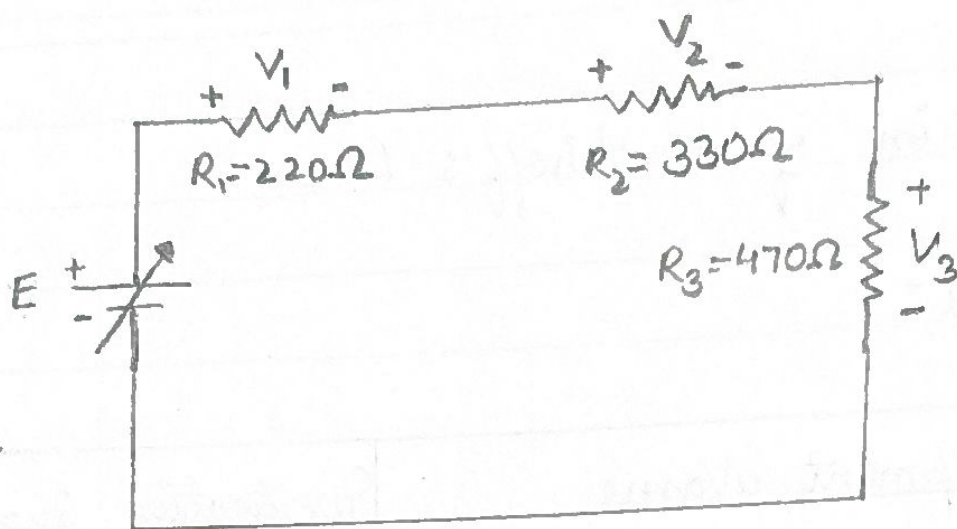


Fig-2: Circuit diagram for KVL

Observation Table -

→ Kirchhoff's Current Law

S.NO.	Current in R_1 (I_1 in mA)	Current in R_2 (I_2 in mA)	Current in R_3 (I_3 in mA)	$I_1 = I_2 + I_3$ (mA)
1.	20.80	14.20	6.60	20.80

Kirchhoff's Voltage Law (KVL):

Kirchhoff's second law is based on the principle of conservation of energy.

It states that the algebraic sum of all voltages in a mesh (loop) in a network at any instant is zero.

KVL can be expressed mathematically as $\sum V = 0$

Consider Fig. 2, applying KVL in loop

$$E - V_1 - V_2 - V_3 = 0$$

$$E = V_1 + V_2 + V_3$$

Procedure:

Kirchhoff's Current Law

- (i) Connect the different instruments and equipment as per circuit diagram.
- (ii) Set the voltage source to minimum position by rotating the knob in anticlockwise direction.
- (iii) Switch ON supply.
- (iv) Increase the source voltage and note down the reading of currents A_1 , A_2 & A_3 by the given ammeter.
- (v) Repeat step 4 for different settings of source voltage.
- (vi) Repeat step 4 and step 5 by constructing circuits with different values: R_1 , R_2 and R_3 .

Observation Table —

→ Kirchhoff's Voltage Law

S.No.	Supply Voltage (E)	Voltage across $R_1(V_1)$	Voltage across $R_2(V_2)$	Voltage across $R_3(V_3)$	V ($V_1+V_2+V_3$)	Loop Voltage ($E-V$)
1.	10	2.18	3.30	4.65	10.13	$10 - 10.13$ $= -0.13V$

Kirchhoff's Voltage Law :

- (i) Connect the different instruments and equipment as per circuit diagram.
- (ii) Set the voltage source to minimum position by rotating the knob in anticlockwise direction.
- (iii) Switch ON supply.
- (iv) Increase the source voltage and note down the reading across E , R_1 , R_2 & R_3 .
- (v) Repeat step 4 for different settings of source voltage.
- (vi) Note the readings of E , VR_1 , VR_2 & VR_3 by the given single voltmeter and put it in the observation table as shown below.

Precautions :

- (1) All the connections should be tight.
- (2) All the steps should be followed carefully.
- (3) Reading should be taken carefully.
- (4) Do not touch the live terminals.